

Chinyere Gloria Mbaka

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Professional Profile

Ph.D. candidate in Electrical and Computer Engineering with expertise in real-time embedded systems, embedded AI, and exposure science instrumentation. Research focuses on the design, calibration, validation, and deployment of Wireless Personal Use Monitors (wPUMs) for measuring puff topography in natural environments. Contributing to NIH-funded research in tobacco regulatory science, with applications in behavior-based exposure modeling and public health assessment. Her work supports the development of reliable monitoring systems for exposure assessment and regulatory decision-making in the United States.

Research Interests

Embedded systems, real-time systems, embedded AI, exposure science instrumentation, sensor calibration, data analysis, machine intelligence, wireless monitoring systems, regulatory science applications.

Education

Rochester Institute of Technology, USA

Aug 2023 – Present

Ph.D., Electrical and Computer Engineering

GPA: 3.62/4.0

Selected Graduate Coursework: Analytical Topics in Computer Engineering; Research Methods in Electrical and Computer Engineering; Interdisciplinary Research Methods; Applied Programming in C; Interface and Digital Electronics; Real-Time and Embedded Systems; Machine Intelligence; Deep Learning; Digital Integrated Circuit Design; Engineering Analysis; Human Factors in AI; Translating Discovery into Practice.

Obafemi Awolowo University, Nigeria

2019 – 2021

M.Tech, Electronics and Telecommunication Engineering

Distinction

Thesis: *Duct Propagation: Aiding Naval Communication at Sea*

Federal University of Technology Owerri, Nigeria

2002 – 2007

B.Eng., Electrical and Electronic Engineering

GPA: 3.99/5

Best Female Graduating Student

Research Experience

Research Assistant

Rochester Institute of Technology

Aug 2023 – Present

- Design, development, and validation of Wireless Personal Use Monitors (wPUMs)
- Calibration using programmable emissions systems (PES) and Alicat flow meters
- Analysis of puff topography parameters including flow rate, duration, volume, and interpuff interval

- Supported multi-phase observational studies (OS4–OS8) involving real-world deployment and large-scale data collection
- E-cigarette emissions testing and product characterization
- Contributing author to manuscript on Gen 3 monitor validation and performance evaluation
- Contributed to data-driven evaluation of monitor performance for exposure science applications

Publications and Presentations

Hyde Electronic Cigarette Aerosol Emissions Characterization

Nov 2024

Air Pollution and Health Research Symposium, Rochester, NY

- Conducted laboratory emissions testing using structured puffing profiles
- Characterized aerosol yield and operating envelope
- Quantified total particulate matter and exposure metrics
- Provided data supporting exposure estimation and behavior-based yield modeling

Ambulatory Puff Topography of Participants in a Natural Environment Study

2025

AAAR Annual Conference

- Investigated real-world puff behavior using wearable embedded monitors
- Identified cumulative puff volume as a key determinant of consumption
- Contributed to understanding variability in real-world user behavior

Manuscripts in Preparation

Validation and Performance Evaluation of Generation 3 wPUM Monitors

- Comprehensive evaluation of accuracy, repeatability, and calibration stability
- Based on more than 160 real-world deployments
- Supports exposure science and regulatory applications
- Intended for submission to a peer-reviewed journal

Professional Service

Abstract Reviewer

NIH Tobacco Regulatory Science Meeting

2026

- Selected to review scientific abstracts for a national-level research meeting
- Evaluated research quality, methodology, and regulatory relevance
- Contributed to the review of a large pool of submissions across the conference

Professional Experience

Weapon Electrical Officer (Lieutenant Commander)

Nigerian Navy

2012 – 2023

- Managed electrical, electronic, and control systems on naval platforms
- Led maintenance and operational readiness of mission-critical systems
- Supervised engineering teams in high-reliability environments
- Provided technical leadership in complex operational settings
- Final role: Staff Officer II at Naval Headquarters

Professional Memberships

- Member, IEEE (Institute of Electrical and Electronics Engineers)
- IEEE Consultants Network
- IEEE Women in Engineering (WIE)
- IEEE Young Professionals
- Member, Nigerian Society of Engineers (NSE)
- Member, Council for the Regulation of Engineering in Nigeria (COREN)

Honors and Awards

- **Chief of the Naval Staff Award for Best Overall Graduating Student**
Ranked 1st among all graduating officers across the Army, Navy, Air Force, and allied participants
Junior Command and Staff Course, Armed Forces Command and Staff College (AFCSC), Jaji, Nigeria
2021
- **Director's Award for Best Overall Graduating Student**
Junior Command and Staff Course, Department of Maritime Warfare, Armed Forces Command and
Staff College (AFCSC), Jaji, Nigeria
2021
- **Director's Award for Leadership**
Junior Command and Staff Course, Department of Maritime Warfare, Armed Forces Command and
Staff College (AFCSC), Jaji, Nigeria
2021
- **Commandant's Exemplary Leadership Award**
Nigerian Navy Engineering College (NNEC), Sapele, Nigeria
2018
- **Best Female Graduating Student in Electrical/Electronic Engineering**
Federal University of Technology Owerri, Imo State, Nigeria
2007

Technical Training

- Nigerian Offshore Patrol Vessel Technical Training, China Shipbuilding Industry Corporation (China)
- Engineering Watchkeeping Certification
- NATO Civil-Military Training (Sweden)

- Maritime Security Training (Ghana)
- Officers Application Course, Nigerian Navy Engineering College

Technical Skills

Programming: C, C++, Python, MATLAB, Embedded C, Assembly

Systems: Embedded Systems, Real-Time Systems, Embedded AI, Control Systems

Tools: STM32, FreeRTOS, MATLAB, Simulink, Keil uVision, CubeMX, Git